

Functional & Performance Testing Template
Model Performance Test

Date	12 july 2026
Team id	
Project Name	Smart Agriculture
Maximum marks	4 marks

Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	User Registration Validation	Register using valid and invalid details (name, email, password)	Valid registration succeeds; invalid inputs display error messages	Application behaved as expected	Pass
FT-02	Login Authentication	Login with correct and incorrect credentials	Valid credentials allow login; invalid credentials show an error	Login functionality worked correctly	Pass
FT-03	Soil Data Input Validation	Enter valid and invalid soil parameters (pH, moisture, NPK values)	Valid inputs accepted; invalid values rejected with validation message	Validation worked correctly	Pass
FT-04	AI Crop Recommendation	Submit soil and weather data and click Predict	System recommends the most suitable crop	Correct crop recommendations displayed	Pass
FT-05	Weather Forecast Integration	Fetch weather information for a selected location	Current weather and forecast displayed accurately	Weather data retrieved successfully	Pass
FT-06	Crop Disease Detection	Upload a crop leaf image for disease detection	AI identifies disease and suggests treatment	Disease detected successfully	Pass
FT-07	Fertilizer Recommendation	Provide crop and soil details	Suitable fertilizer recommendations are displayed	Recommendations generated correctly	Pass
FT-08	Notification System	Trigger weather alert or disease alert	User receives timely notifications	Notifications delivered successfully	Pass

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
PT-01	Response Time Test	Measure crop recommendation response time	Response generated within 3 seconds	Average response time: 2.4 seconds	Pass
PT-02	Concurrent User Test	Simulate multiple users requesting predictions simultaneously	System handles multiple requests without delays	Stable performance observed	Pass
PT-03	Image Upload Performance	Upload multiple crop images for disease detection	Images processed without crashing or errors	All images processed successfully	Pass
PT-04	Dashboard Load Test	Open dashboard with multiple datasets	Dashboard loads smoothly within acceptable time	Loaded in 2.8 seconds	Pass
PT-05	Database Performance	Save and retrieve farmer records repeatedly	Data stored and retrieved without errors	Database operations completed successfully	Pass